Action Archive

for BigFix Platform

Administrator Guide

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Platform	Ubuntu Server 22.04 / 24.04 LTS
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1. Introduction

1.1 Purpose of This Guide

This guide provides comprehensive reference and procedural documentation for administrators responsible for deploying, configuring, and maintaining the Action Archive for BigFix Platform application. It covers all administrative functions available through the web interface as well as server-side operations performed via the Linux command line.

1.2 Intended Audience

This guide is written for:

- BigFix administrators who manage day-to-day operations and will serve as the primary users of the application
- IT operations staff who maintain the underlying server infrastructure
- Security and compliance personnel who require audit visibility into BigFix action history

Readers are assumed to be comfortable with BigFix concepts (operators, fixlets, actions, sites), Linux system administration, and general networking. BigFix itself is not described in this guide.

1.3 Document Conventions

Convention	Meaning
ℹ️ NOTE	Additional context, clarification, or technical background
▶️ TIP	A helpful suggestion or best practice
❗ IMPORTANT	A critical requirement that must be understood before proceeding
⚠️ WARNING	A condition or action that may result in data loss or a security risk
✘ CRITICAL	Severe risk; take immediate action before proceeding
Monospace / COMMAND	Terminal commands, file paths, configuration values, and code samples

1.4 Application Overview

Action Archive for BigFix Platform is an community-created web application that captures, preserves, and presents stopped and expired actions visible in the BigFix Console. Because the BigFix Console does not retain detailed records of actions once deleted, Action Archive fills this gap by periodically querying the BigFix REST API and storing full action metadata in a local PostgreSQL database so they can safely be deleted from the console without losing historical visibility.

Key value propositions:

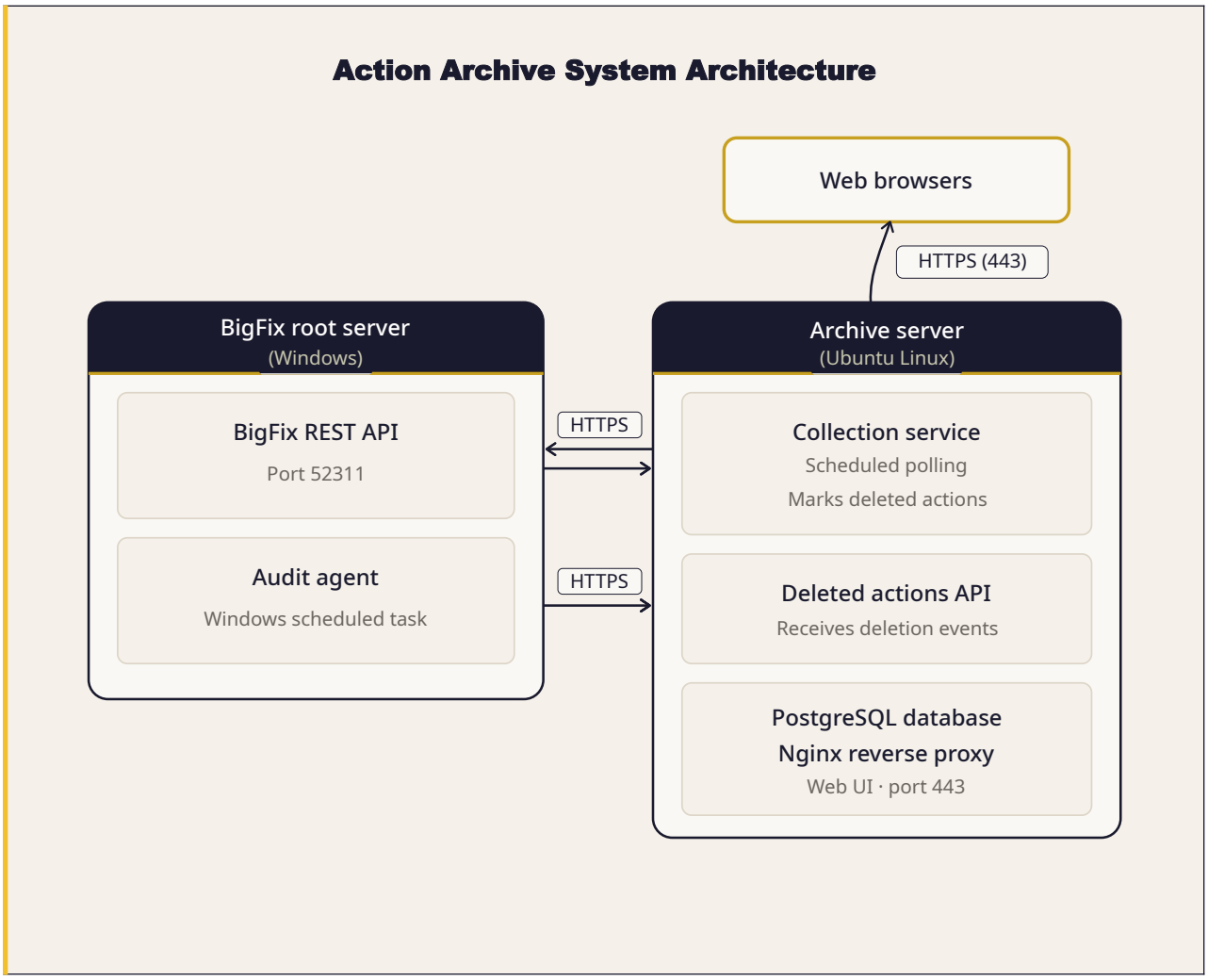
- Compliance and audit readiness: a permanent, searchable record of every BigFix action and its per-device results
- Operational visibility: instant answers to "Who deployed this patch?" and "Which devices failed?"
- Deletion tracking: the Audit Agent captures action deletions that would otherwise leave no record in the BigFix Console.
- Role-based access: separate admin, auditor, and viewer roles enforce least-privilege access
- Export capability: action lists and audit logs exportable to CSV and PDF for compliance reporting

> ℹ️ **NOTE** Action Archive collects expired and stopped actions only. It is not a real-time monitoring tool and does not interfere with BigFix operations in any way.

2. Architecture Overview

2.1 System Architecture Diagram

The following diagram illustrates the major components of the Action Archive deployment and the data flows between them.



2.2 Component Descriptions

Component	Technology	Role
Archive Server	Ubuntu Server 22.04 / 24.04 LTS	Host VM for all application services
Reverse Proxy	Nginx	Terminates TLS, enforces security headers, proxies to FastAPI on localhost:5000
Application Server	Python 3.12 / FastAPI / Uvicorn	Handles all API requests, collection logic, authentication, and business rules
Database	PostgreSQL 16	Stores archived actions, users, settings, audit logs, and collection history
BigFix Root Server	HCL BigFix Platform	Source of action data; queried via REST API (port 52311) and Session Relevance
Audit Agent	Python / Windows Scheduled Task	Monitors server_audit.log on BigFix server; reports action deletions to Archive API
Browser Clients	Google Chrome, Mozilla Firefox, Microsoft Edge	Access the Action Archive web UI over HTTPS (port 443)

2.3 Network Requirements

Port	Protocol	Direction	Purpose
443	TCP / HTTPS	Inbound to Archive Server	Web UI and API access from browsers and the Audit Agent
52311	TCP / HTTPS	Outbound from Archive Server	BigFix REST API and Session Relevance queries
5000	TCP / HTTP	Localhost only	Internal: Nginx → FastAPI (not exposed externally)
5432	TCP	Localhost only	Internal: FastAPI → PostgreSQL (not exposed externally)
25/587/465	TCP / SMTP	Outbound from Archive Server	Email notifications and password reset (if SMTP configured)

❗ **IMPORTANT** Ports 5000 and 5432 are bound to localhost only and must never be exposed externally. All external access occurs exclusively through Nginx on port 443.

2.4 Deployment Model

Action Archive is a single-instance application. Only one BigFix Root Server is supported for each Archive Server. There is no clustering or high-availability configuration. Redundancy is achieved through regular database backups and VM-level snapshots of the host.

The application is designed to support several concurrent users. Storage growth is estimated at approximately 15 MB per 1,000 actions.

3. Getting Started — First Login

3.1 Accessing the Web Interface

Open a web browser and navigate to:

```
COMMAND
https://<server-ip>
  or
https://<server-hostname>
```

3.2 Self-Signed Certificate Warning

By default, Action Archive uses a self-signed SSL certificate. Your browser will display a security warning on first access. This is expected behavior in a standard deployment.

To proceed through the warning:

1. Click Advanced or More Information.
2. Click Proceed to <address> or Accept the Risk and Continue.
3. The login page will load normally. The warning will not reappear in the same browser.

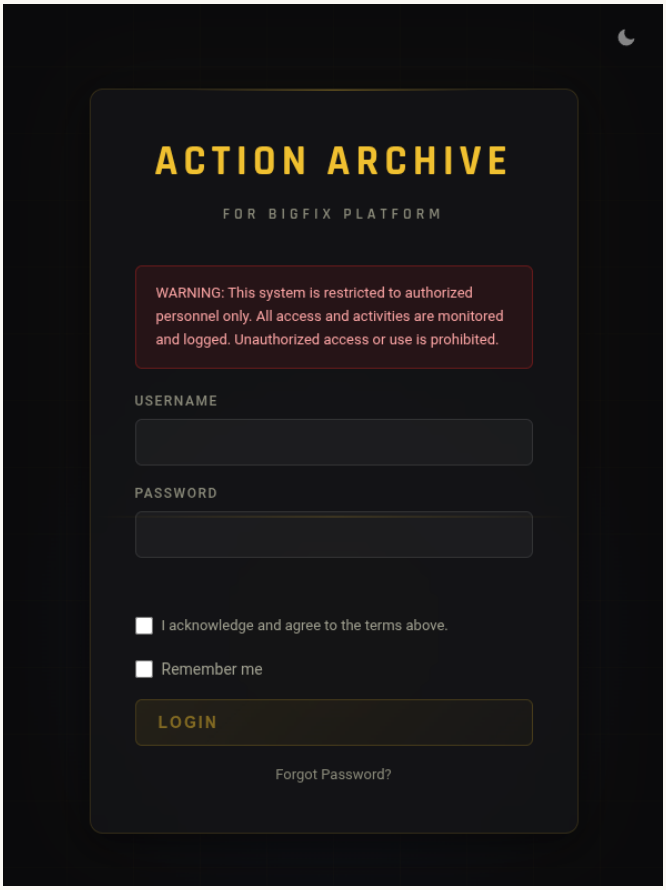
> ℹ️ **NOTE** To eliminate the browser warning permanently, replace the self-signed certificate with one signed by your internal Certificate Authority. See Section 9.5 (Certificates) for instructions.

3.3 Login Page

The login page provides the following interactive features:

- Theme toggle: Switch between Dark and Light themes. Your selection persists across sessions.
- **[If Configured]:** Use Consent checkbox: Must be checked before login is permitted. Confirms acknowledgment of acceptable-use terms.
- **[If Configured]:** Remember Me: Extends the session token lifetime.

Enter the administrator credentials supplied during installation, then click Sign In.



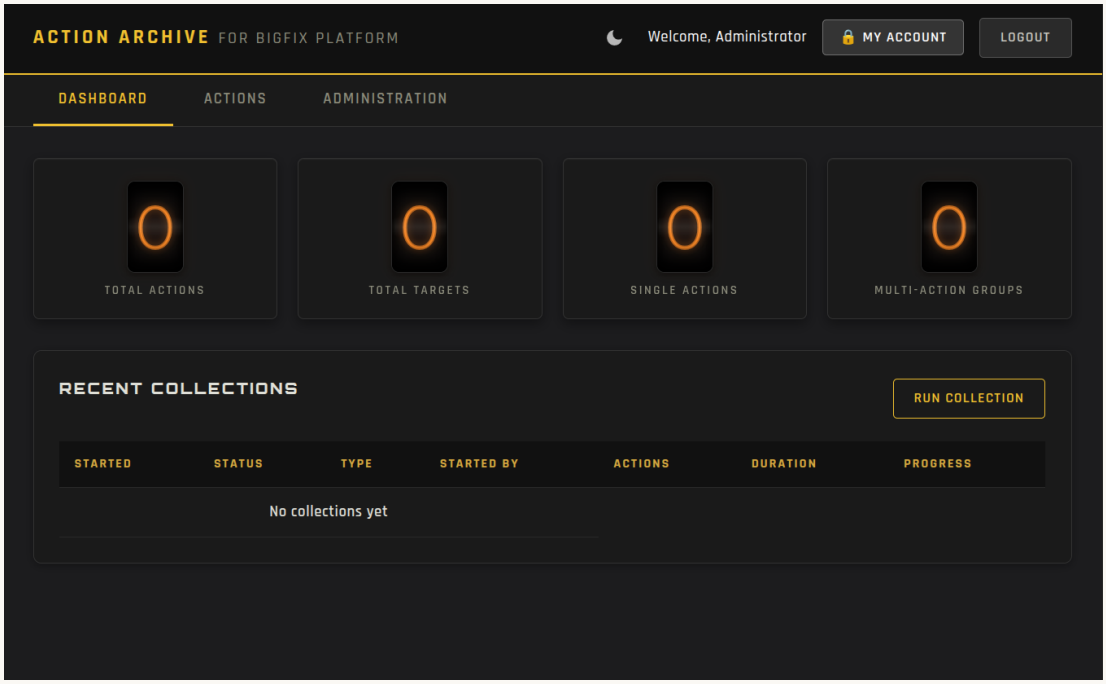
3.4 Default Credentials

> ❗ **IMPORTANT** Credentials are set during the installation process. There are no factory-default username or password values hard-coded in the application.

On first login, if the Must Change Password flag is set on your account, you will be prompted to choose a new password before reaching the dashboard.

3.5 First-Time Dashboard View

If no collection has been run, the dashboard will display zero counts across all stat cards and an empty Recent Collections table. To populate the archive, click Collect Now (see Section 4.3) or wait for a scheduled collection.



4. Dashboard

4.1 Stat Cards

The top of the dashboard displays four stat cards summarizing the current contents of the archive:

Card	Description
Total Actions	Total count of all archived actions, including single actions and Multi-Action Groups.
Total Targets	Cumulative count of device-action assignments across all archived actions.
Single Actions	Count of single actions.
Multi-Action Groups	Count of parent MAG and Baseline records that contain component actions.

4.2 Recent Collections Table

Below the stat cards, the Recent Collections table lists the last several collection runs:

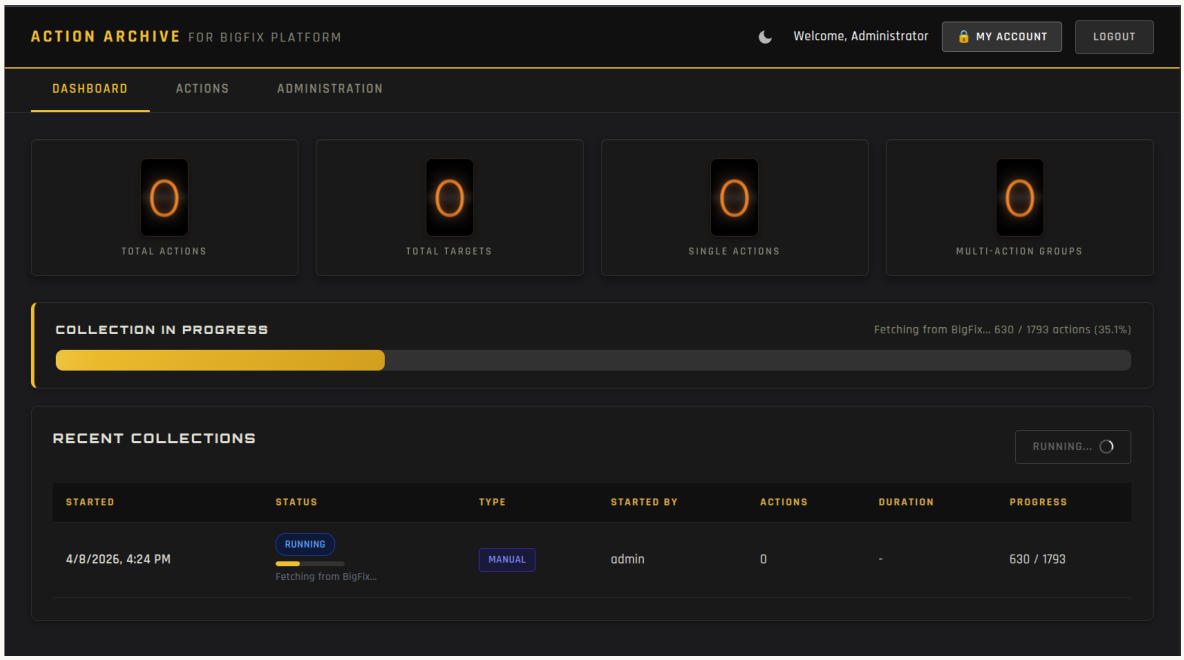
Column	Description
Started	Date and time the collection began (in your local browser timezone).
Duration	Total elapsed time for the collection run.
Actions Found	Number of new actions discovered and archived in this run.
Status	Outcome: Completed, Running, or Error.
Details	Link to the full per-phase collection log for this run.

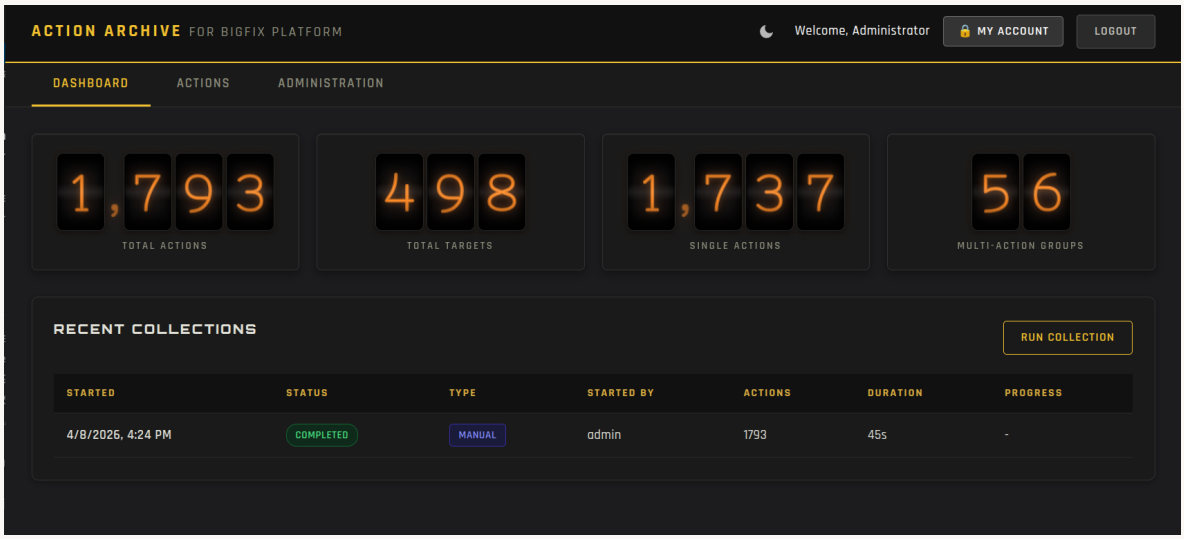
4.3 Run Collection Button

The Run Collection button triggers an immediate, on-demand collection and is available to Admin-role users only. During an active collection, the button is replaced by a live progress bar. If a scheduled collection is already running, the on-demand run is queued to start immediately after the current one completes.

4.4 Live Collection Progress

While a collection is in progress, a progress bar displays the current phase name (for example, Fetching or Storing) and a percentage completion. The bar disappears automatically on completion, and the stat cards reflect the updated counts.

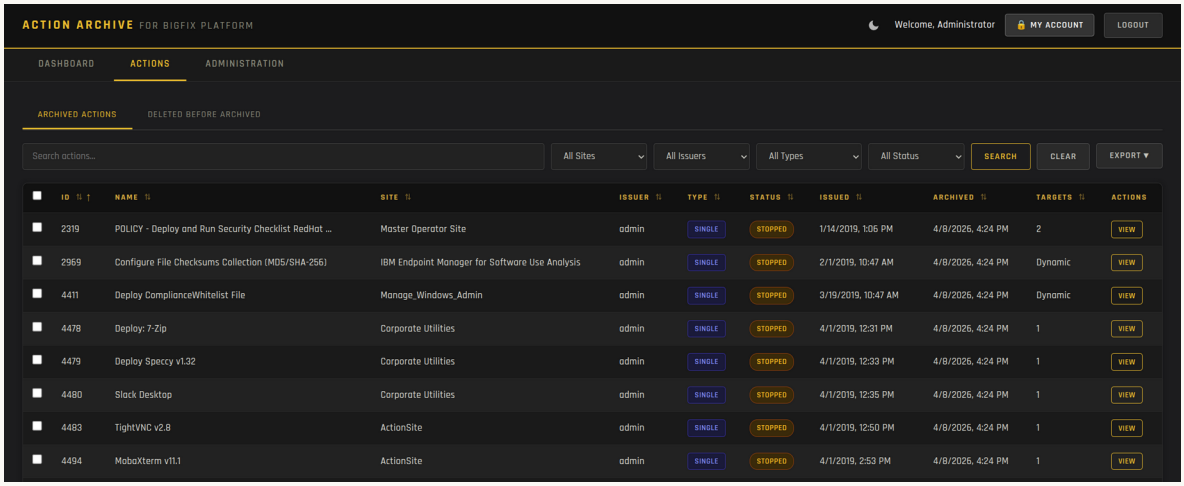

ACTION ARCHIVE FOR BIGFIX PLATFORM
Welcome, Administrator
MY ACCOUNT
LOGOUT
DASHBOARD
ACTIONS
ADMINISTRATION
0
TOTAL ACTIONS
0
TOTAL TARGETS
0
SINGLE ACTIONS
0
MULTI-ACTION GROUPS
COLLECTION IN PROGRESS
Fetching from BigFix... 630 / 1793 actions (35.1%)
RECENT COLLECTIONS
RUNNING...
STARTED
STATUS
TYPE
STARTED BY
ACTIONS
DURATION
PROGRESS
4/8/2026, 4:24 PM
RUNNING
Fetching from BigFix...
MANUAL
admin
0
-
630 / 1793


ACTION ARCHIVE FOR BIGFIX PLATFORM
Welcome, Administrator
MY ACCOUNT
LOGOUT
DASHBOARD
ACTIONS
ADMINISTRATION
1,793
TOTAL ACTIONS
498
TOTAL TARGETS
1,737
SINGLE ACTIONS
56
MULTI-ACTION GROUPS
RECENT COLLECTIONS
RUN COLLECTION
STARTED
STATUS
TYPE
STARTED BY
ACTIONS
DURATION
PROGRESS
4/8/2026, 4:24 PM
COMPLETED
MANUAL
admin
1793
45s
-

5. Browsing and Searching Actions

5.1 Actions Tab Overview

The Actions tab provides the primary interface for exploring archived actions. It opens to a paginated table listing all archived actions, with controls for searching, filtering, sorting, and exporting.



5.2 Column Descriptions

Column	Description	Notes
ID	BigFix-assigned numeric action identifier.	
Name	Display name as defined in BigFix.	—
Type	Single Action or Multi-Action Group (MAG / Baseline).	—
Site	Fixlet site the source fixlet belongs to.	May show N/A if source fixlet was deleted.
Issuer	BigFix operator who created the action.	—
Status	Last known action status (Expired, Stopped, etc.).	—
Issued	Date and time the action was created in BigFix.	Displayed in your local browser

		timezone.
Targets	Number of devices the action was targeted at.	—
Archived	Date and time the action was collected into the archive.	
Actions	View button (click to see a detail of the action)	

5.3 Searching and Filtering

The search bar accepts free-text queries and matches against Action ID, Action Name, Issuer, and Site simultaneously. Additional filter controls:

- Site: Restrict results to actions from a specific content site.
- Issuer: Filter to actions created by a specific BigFix operator.
- Type: Show only Single Actions or only Multi-Action Groups.
- Status: Filter by action status (Expired, Stopped, etc.).

All filters are additive. Click Reset Filters to clear all active selections.

5.4 Sorting and Pagination

Click any column header to sort by that field; click again to reverse the sort order. The active sort is indicated by an arrow icon in the column header. Use the pagination controls at the bottom to navigate pages, and the page-size selector to change the number of rows displayed per page.

5.5 Action Detail View

Click the “View” button under the “Actions” column to open the detail view of the specific action. Action details include:

- Core properties: name, type, site, issuer, status, dates, Action Script and Relevance expression.
- Components: For Multi-Action Groups and Baselines, lists all member actions with individual IDs, names, and statuses. BigFix copies fixlet content inline when deploying baselines, so component records reflect content at deployment time.
- Targets: Every device the action was targeted at, with per-device status, last report time, and result message.

REMOVE BIGFIX SELF-SERVICE APPLICATION

ACTION INFORMATION

ACTION ID	TYPE	STATUS	SITE
18944	SINGLE	STOPPED	Software Distribution

ISSUER	ARCHIVED
mconsuegra	4/6/2026, 5:18 PM

ACTION TIMELINE

ISSUED
4/6/2026, 4:50 PM

TARGET SUMMARY

1 TOTAL	0 COMPLETED	1 FAILED	0 PENDING	0 NOT RELEVANT	0 NOT REPORTED

RELEVANCE

```
(((if exists property "in proxy agent context" then ( not in proxy agent context ) else true )AND (if
exists property "android" of type "operating system" then ( not android of operating system ) else true
)) AND ((if exists property "in proxy agent context" then (not in proxy agent context) else true))) AND
((if (windows of operating system) then (exists key whose ((it as string = "IBM BigFix Self Service
Application" OR it as string = "IBM BigFix Self-Service Application" OR it as string = "BigFix Self-
Service Application") of value "DisplayName" of it) of key
"HKLM\Software\Microsoft\Windows\CurrentVersion\Uninstall" of x32 registry) else if (mac of operating
system AND exists file "/Library/Application Support/BigFix/BigFixSSA/ssa.config") then ((exists
application (percent decode(following text of first "applicationName: " of lines of file
"/Library/Application Support/BigFix/BigFixSSA/ssa.config" as trimmed string |"BigFixSSA") & ".app"))
OR (exists application "BigFixSSA.app")) else false))
```

ACTION SCRIPT

```
setting delete "_BESClient_ActionManager_SelfServiceApp" on "{now}" for client
setting delete "_BESClient_ActionManager_SSAv2Mode" on "{now}" for client
setting delete "_BESClient_ActionManager_SSAv3Enable" on "{now}" for client

if {exists setting "_BESClient_ActionManager_HideProgressTab_backup" whose (value of it = "0") of
client}
        setting "_BESClient_ActionManager_HideProgressTab"="0" on "{now}" for client
        setting delete "_BESClient_ActionManager_HideProgressTab_backup" on "{now}" for client
else
        setting "_BESClient_ActionManager_HideProgressTab"="0" on "{now}" for client
endif

if {(exists setting "_BESClient_ActionManager_TrayMode_backup" whose (value of it as lowercase !=
"notray") of client) AND (exists setting "_BESClient_ActionManager_TrayMode" whose (value of it as
lowercase = "notray") of client)}
```

TARGET COMPUTERS (1)

Export Targets

COMPUTER	STATUS	STARTED	COMPLETED
BFROOTSVR	FAILED	4/6/2026, 4:52 PM	4/6/2026, 4:52 PM

This action was deleted from the BigFix console on 4/7/2026, 11:28 AM

6. Deleted Actions Tracking

6.1 Overview

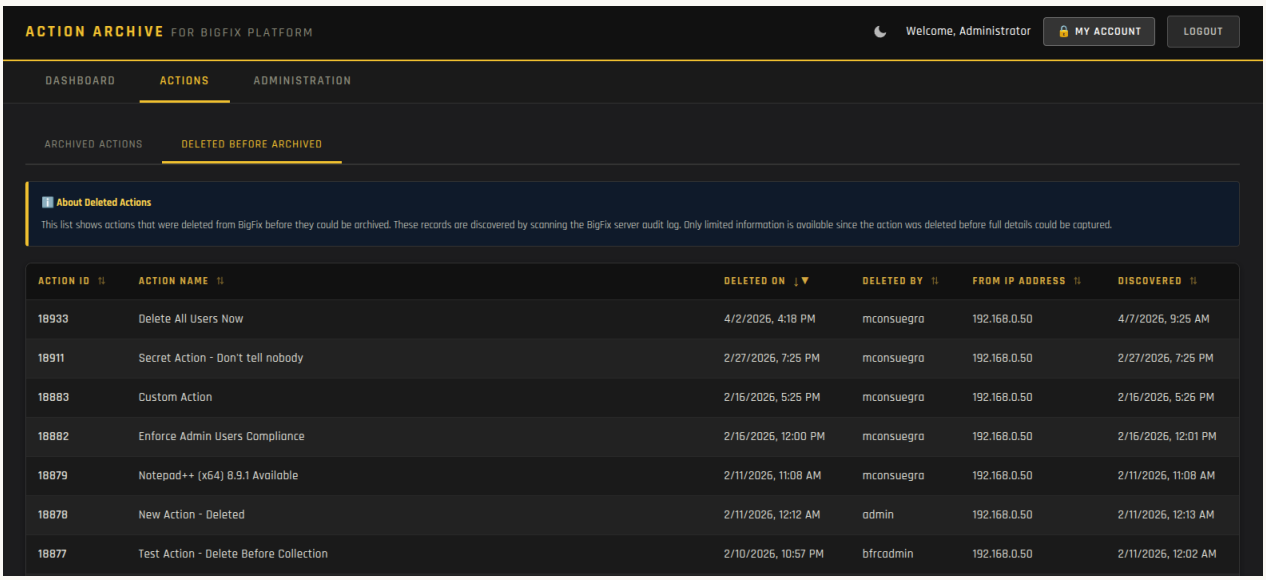
The Deleted Before Archived sub-tab within the Actions tab lists BigFix actions that were deleted from the BigFix server before Action Archive had the opportunity to collect them. This provides visibility into a gap inherent to any periodic collection approach: actions created and deleted between collection cycles.

6.2 How Data Gets Here

The Audit Agent (see Section 11) monitors the BigFix server_audit.log file using a Windows Scheduled Task. When the agent detects an action deletion event, it submits that event to the Archive REST API using an API key. The record is stored even if the action's full details were never collected.

6.3 Fields Displayed

Field	Description
Action ID	BigFix action identifier as recorded in the audit log.
Action Name	Action name from the audit log, if available.
Deleted On	Timestamp when the deletion was recorded in server_audit.log.
Deleted By	BigFix operator username that performed the deletion.
From IP Address	IP address of the BigFix Console from which the deletion was performed.
Discovered	Indicates when the action was archived.



7. Exporting Data

Action Archive supports several export formats from different parts of the interface. All exports reflect the current filter state — apply filters before exporting to scope the output.

Export	Format	Contents	Accessed From
Action List	CSV	Filtered action metadata: ID, name, type, site, issuer, status, issued date, target count.	Actions tab → Export button
Action List	PDF	Formatted action summary with statistics header and full action table.	Actions tab → Export button
Target Computers	CSV	Per-device status for one action: device name, status, last report, result message.	Action detail → Targets tab
Audit Log	CSV	Full security event log: timestamp, user, event type, details, IP, user agent.	Administration → Logs → Audit Log
Audit Log	PDF	Formatted audit log with statistics header.	Administration → Logs → Audit Log

ℹ️ **NOTE** All exports honor the data-visibility scopes assigned to the current user. A Viewer-scoped user will only export records they are permitted to see.

8. Administration — User Management

User management is available from the Administration menu.

8.1 User Accounts

Creating a User

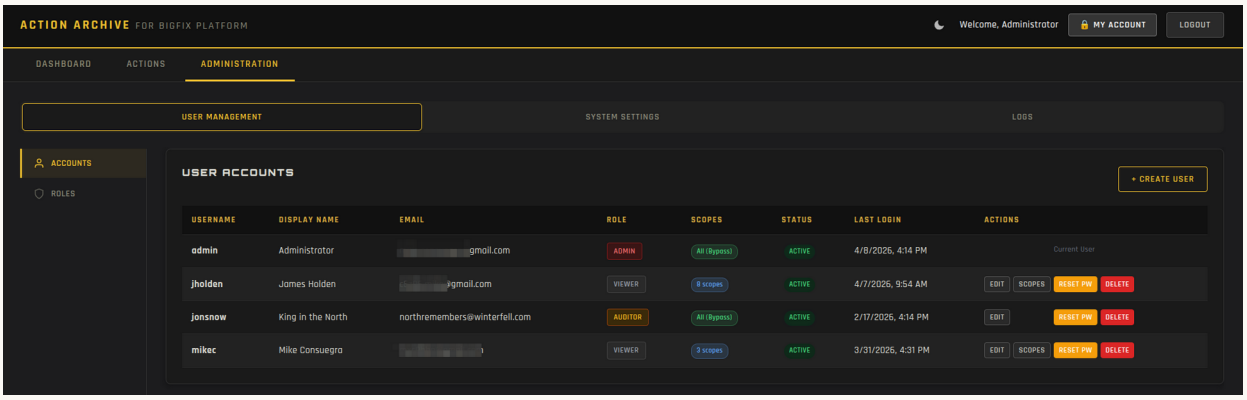
1. Navigate to Administration → Users.
2. Click Create User.
3. Complete all required fields: Username, Email Address, Display Name, Role, and Initial Password.
4. Optionally enable Must Change Password to force a password change on first login.
5. Click Save. The user can log in immediately.

Editing a User

Click the Edit icon next to any user to modify their email address, display name, active status, or role. Username cannot be changed after creation. Changes take effect immediately.

Deleting a User

Click the Delete icon and confirm. Deletion is permanent. Audit log entries associated with the user are retained with the username preserved in the record even after account deletion.



Admin-Initiated Password Reset

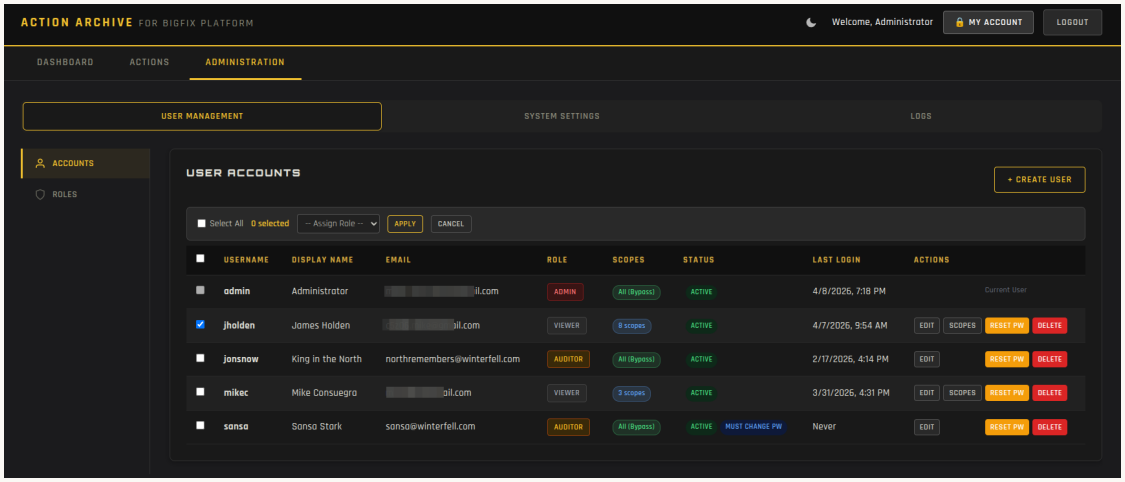
Click Reset Password next to a user account, enter a new temporary password, and save. The 'Must Change Password' flag is automatically set, forcing the user to choose a new password at their next login.

Unlocking a Locked Account

Accounts are locked after the configured number of consecutive failed login attempts (see Section 9.2). Click Unlock next to the affected user to restore access immediately. Accounts also unlock automatically after the configured Lockout Duration.

Bulk Role Assignment

Select multiple users using the checkboxes in the user list, then choose a new role from the Assign Role dropdown to apply it to all selected accounts at once.



8.2 Roles and Permissions

Action Archive uses three roles. The table below details each role:

Capability	Admin	Auditor	Viewer
View archived actions and search	Yes	Yes	Yes
View action detail and target results	Yes	Yes	Yes
Export action lists (CSV / PDF)	Yes	Yes	Yes
View Deleted Before Archived	Yes	Yes	No
View and export Audit Log	Yes	Yes	No
Manage user accounts	Yes	No	No
Change system settings	Yes	No	No
Trigger Collect Now	Yes	No	No
Create and revoke API keys	Yes	No	No
Create and restore backups	Yes	No	No
View application and system logs	Yes	No	No

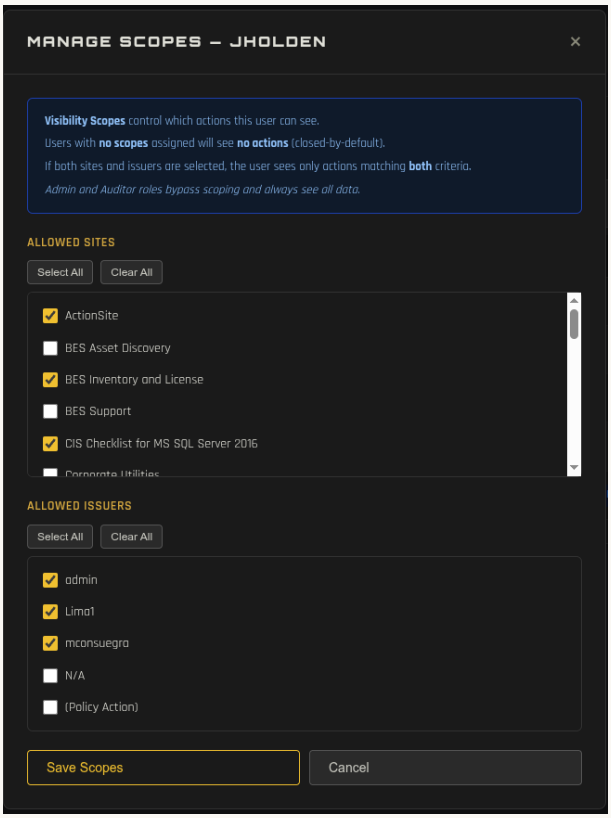
8.3 User Scopes (Data Visibility Restrictions)

Scopes allow administrators to restrict which actions a user can see based on Site or Issuer. A user with no scopes sees all archived actions. A user with scopes assigned sees only actions matching at least one of their scope conditions.

To assign scopes:

1. Open the user's record and click Manage Scopes.
2. Add one or more Site Name or Issuer conditions using the scope editor.
3. Click Save.

Scope enforcement applies to the action list, search results, action detail views, exports, and API calls. Example: members of the Patching team should see only actions from the "Patches for Windows" site. Assign a Site scope of that value to their accounts.



9. Administration — System Settings

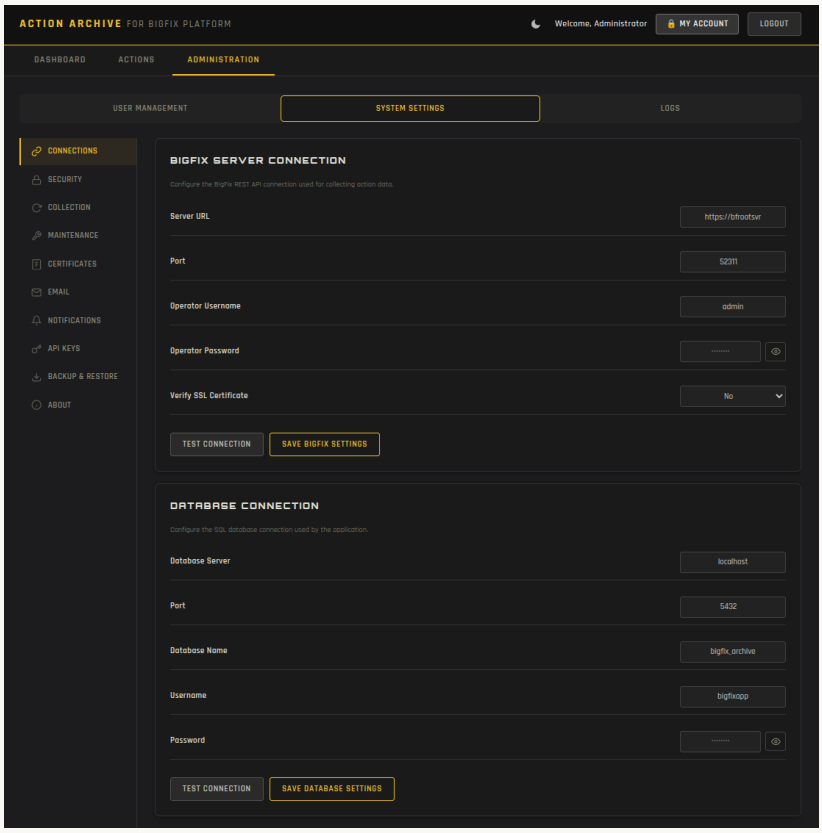
All system settings are accessible to Admin-role users from Administration → Settings. Changes made through the web interface take effect immediately without a service restart unless otherwise noted.

9.1 Connections

Setting	Description	Example / Default
BigFix Server URL	Hostname or IP of your BigFix Root Server.	192.168.1.10
BigFix Port	Port for the BigFix REST API.	52311
BigFix Username	BigFix Master Operator account used for API queries.	
BigFix Password	Password for the above account (encrypted at rest).	
Verify SSL Cert	Validate the BigFix server's SSL certificate. Disable for self-signed BigFix certs.	No (default)
DB Host	PostgreSQL server hostname.	localhost

Setting	Description	Example / Default
DB Port	PostgreSQL listen port.	5432
DB Name	Name of the archive database.	bigfix_archive
DB Username	PostgreSQL user used by the application.	bigfix_user
DB Password	Database password (encrypted at rest).	

❗ **IMPORTANT** The operator account used must have Master Operator (MO) privileges. Without MO access, Session Relevance queries used to discover actions across all sites will return incomplete results.



9.2 Security Settings

Password Policy

Setting	Description	Default
Minimum Length	Minimum number of characters required.	8
Require Uppercase	At least one uppercase letter must be present.	Yes
Require Lowercase	At least one lowercase letter must be present.	Yes
Require Numbers	At least one digit must be present.	Yes
Require Special Chars	At least one special character must be present.	Yes
Password Expiration	Days after which passwords must be changed. 0 = never.	0
Password History Count	Prevents reuse of the last N passwords. 0 = no history.	0

Account Lockout Policy

Setting	Description	Default
Max Failed Attempts	Consecutive failed logins before account is locked.	5
Lockout Duration	Minutes the account remains locked. Admin can unlock manually.	15

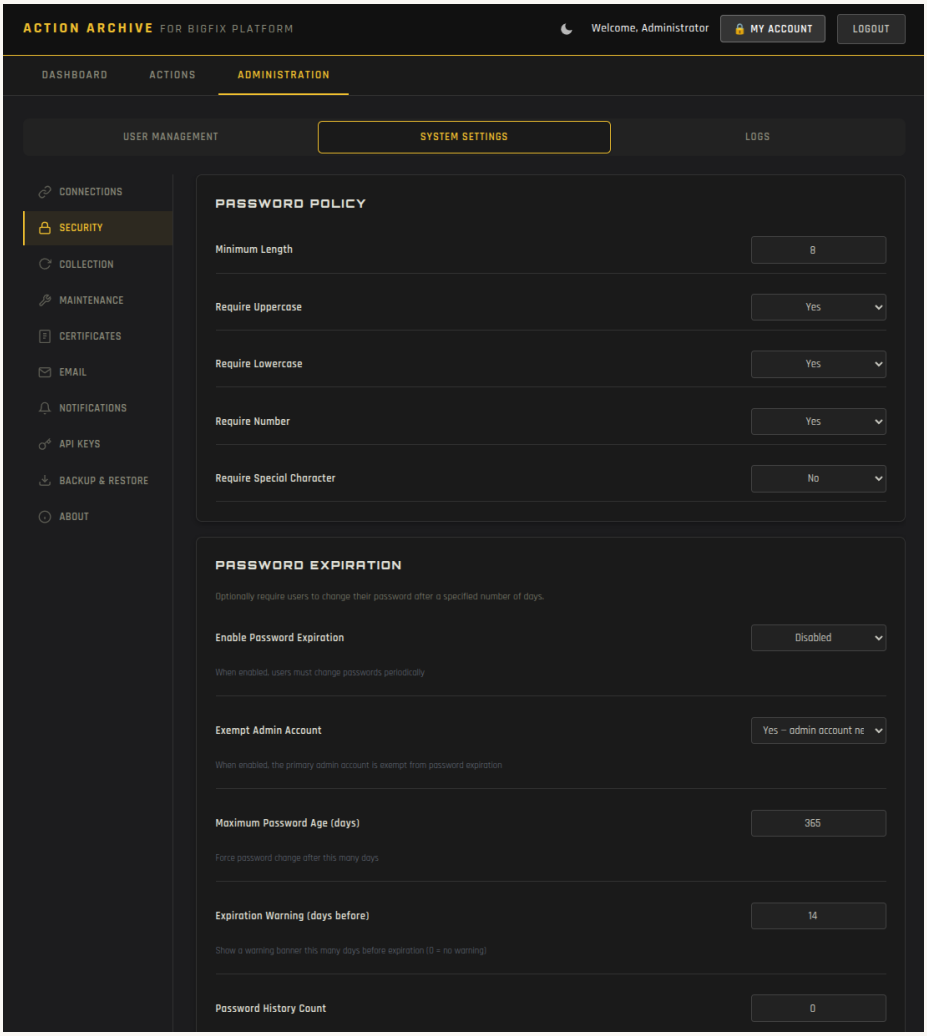
Rate Limiting

Setting	Description	Default
Max Attempts by IP	Number of failed login attempts from a specific IP before it is blocked from attempting to logon.	10
Time Window	Failed attempts during this time window count towards rate limit	15
Ban duration	IP is banned from re-attempting for this time.	15
IP Whitelist	List of IPS not subject to rate limiting	

Login Page Customization

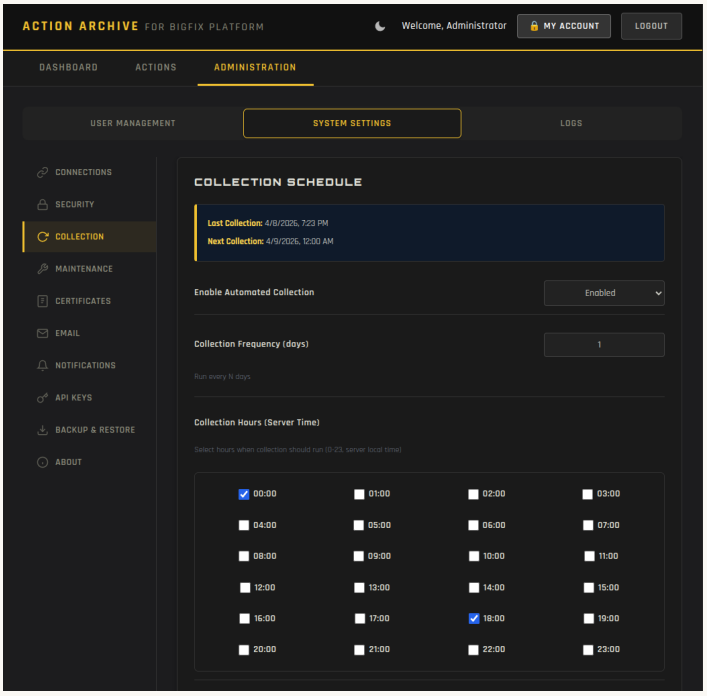
Setting	Description	Default
Banner Message	Message that will display on the logon page if feature is enabled	
Banner Style	Info / Warning / Legal styles	

Setting	Description	Default
Consent Text	Message presented to user for consideration before logging in.	
Require Consent	If enabled, user must acknowledge acceptance of consent message	

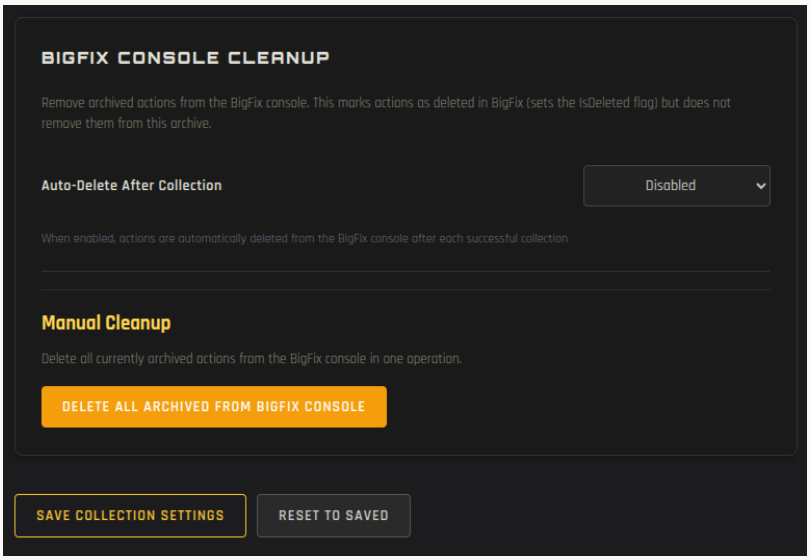


9.3 Collection Schedule

- Enable Automatic Collection: Master toggle to enable or disable scheduled collections.
- Frequency (Every N Days): Run a collection every N days from the last completed run.
- Collection Hours: Specify the hour(s) of the day in which collection will occur.



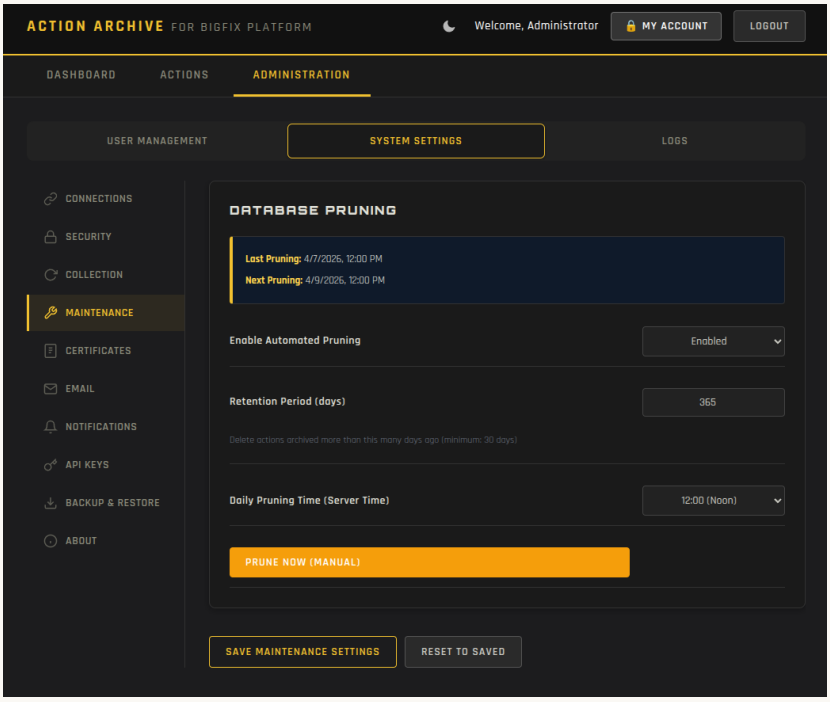
Administrators can also choose to ‘delete’ collected actions from the Console:



9.4 Maintenance (Database Pruning)

- Retention Period (Days): Actions archived more than this many days ago are removed by the pruning job.
- Run Pruning Now: Triggers an immediate pruning operation using the current retention setting.
- Pruning History: A table showing past runs with timestamps and row counts deleted.

> ⚠️ **WARNING** Pruning permanently deletes action records from the archive database. This operation cannot be undone. Ensure your retention period aligns with your organization's data retention policies. Create a database backup (Section 9.9) before running pruning for the first time.



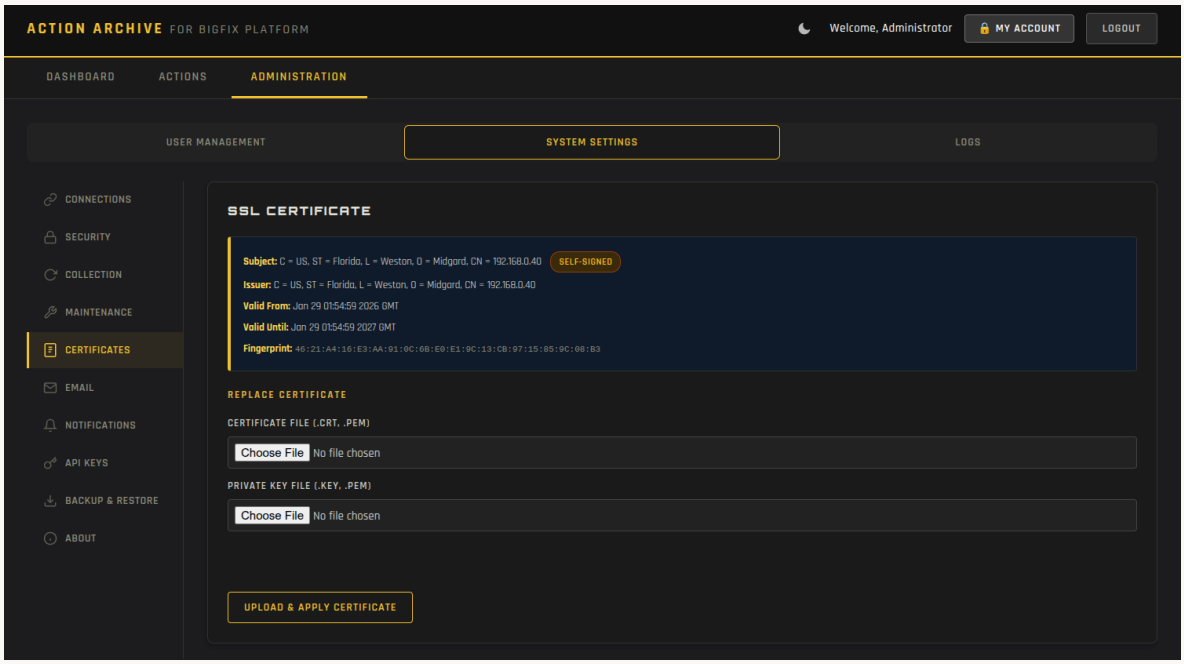
9.5 Certificates (SSL)

This panel displays information about the current SSL certificate used by Nginx and allows you to replace it through the web interface.

- Current Certificate Details: Displays issuer, subject, validity dates, and SHA-256 fingerprint of the active certificate.
- Replace Certificate: Upload a PEM-encoded X.509 certificate chain and corresponding private key. Nginx is reloaded automatically after upload.

> ℹ️ **NOTE** This panel manages the Archive Server's own SSL certificate (what browsers see). It does not affect the BigFix server's certificate. To control BigFix certificate validation, use the Verify SSL toggle in Connections (Section 9.1).

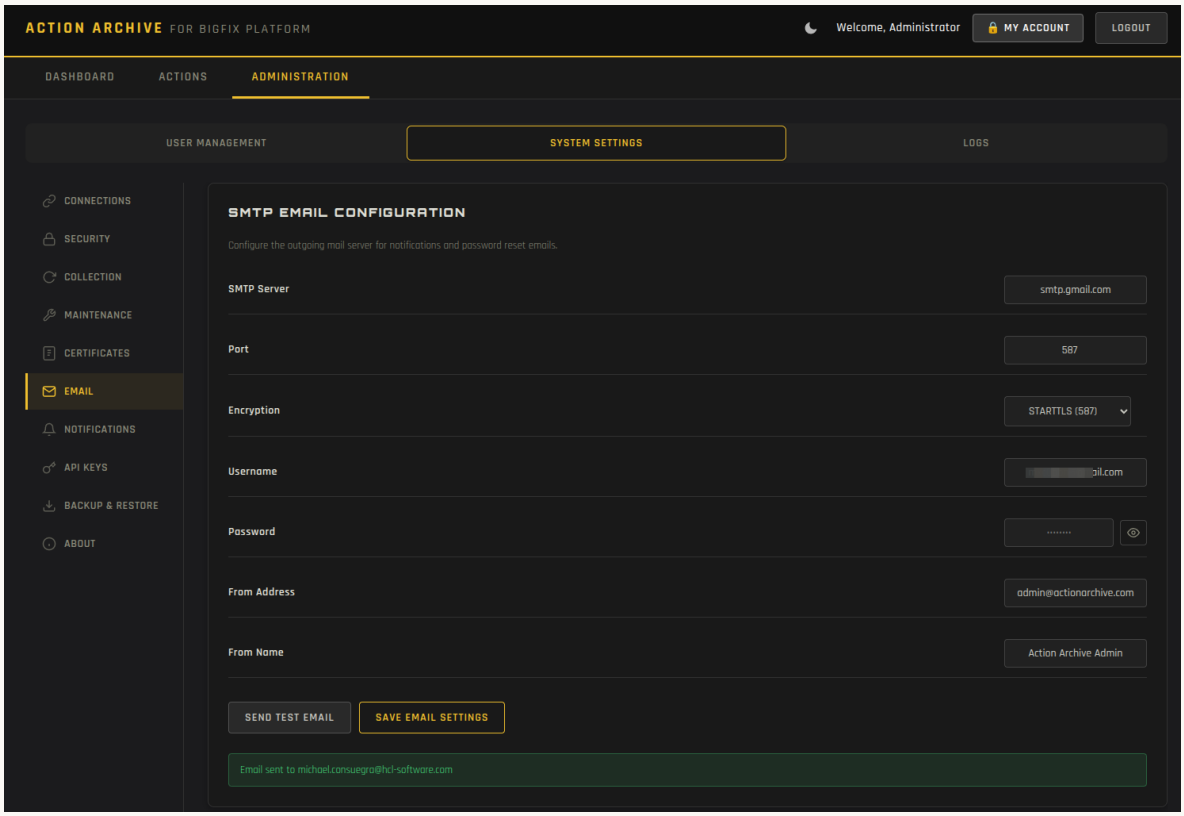
> ❗ **IMPORTANT** The certificate and private key must be in PEM format. The certificate file should include the full chain (server cert + any intermediate CAs), ordered from leaf to root.



9.6 Email / SMTP

Field	Description	Example
SMTP Server	Hostname of your mail relay server.	smtp.example.com
Port	Port for the SMTP connection.	587
Username	SMTP authentication username.	archive@example.com
Password	SMTP password (encrypted at rest).	
Encryption	Connection security: none, starttls, or ssl.	starttls
From Address	Sender address shown on outgoing messages.	archive@example.com
From Name	Sender display name shown in email clients.	Action Archive

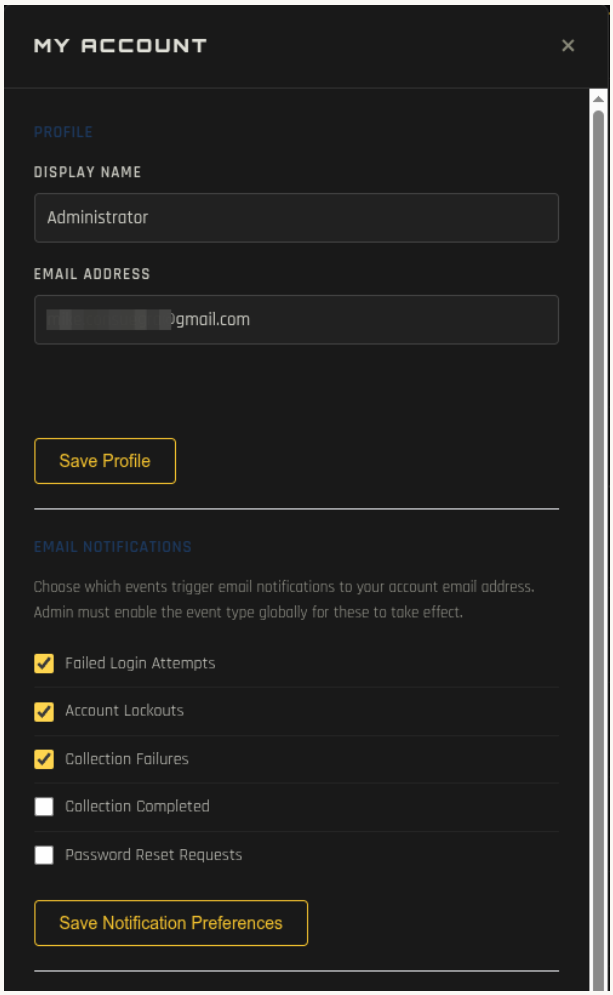
Use the Send Test Email button to verify your SMTP configuration before saving.



9.7 Notifications

Event Type	Description
Completed Collection	Sent after each successful collection run with a summary of actions archived.
Failed Collections	Sent when a collection run encounters an unrecoverable error.
Account Lockouts	Sent when a user account is locked out due to too many invalid login attempts
Password Reset Requests	Confirmation sent to the user when their password is reset.
Failed Login Attempts	Alerts on repeated failed login attempts or account lockouts.

Additional Recipients: A comma-separated list of addresses that receive all enabled notifications in addition to registered users who have opted in. Users can manage individual preferences from the My Account page.



9.8 API Keys

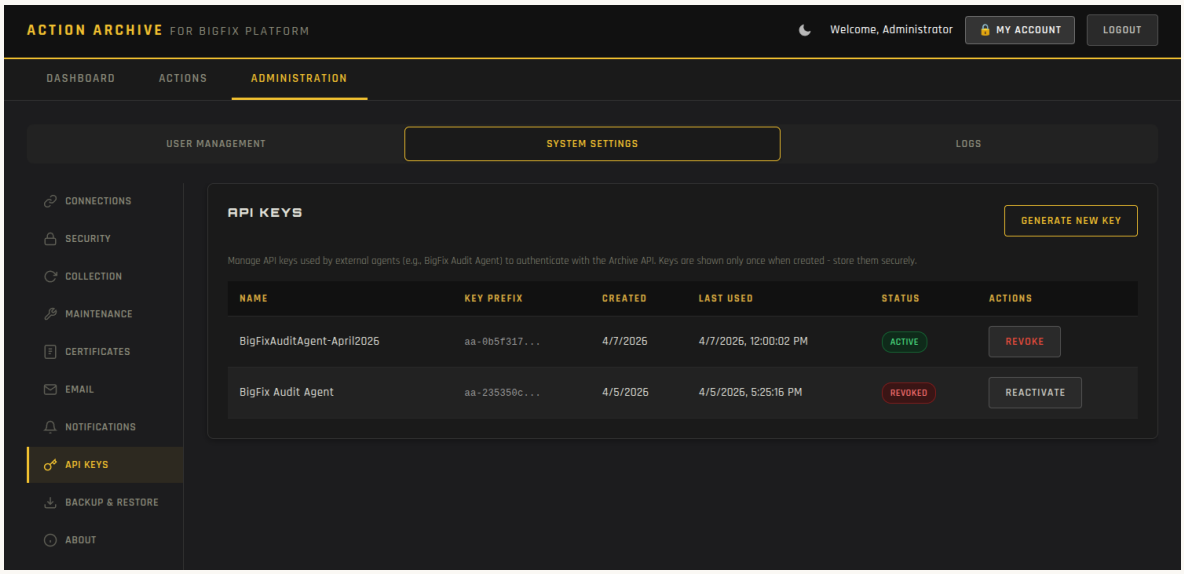
API keys provide machine-to-machine authentication for the Archive REST API without interactive login. The Audit Agent uses an API key to submit discovered deleted actions.

To generate a new API key:

4. Navigate to Administration → Settings → API Keys.
5. Click Generate New Key.
6. Enter a descriptive name (e.g., Audit Agent — BigFix Server 01).
7. Click Generate.
8. Copy the full key from the confirmation dialog immediately.

> ❗ **IMPORTANT** The full API key is displayed only once at creation. The system stores only the key prefix for reference. If the key is lost, revoke it and generate a new one. Store API keys in a secrets manager or protected configuration file.

API requests authenticate by including the key in the X-API-Key HTTP header. Keys can be individually revoked (disabled). Revoked keys remain visible in the list for audit purposes.



9.9 Backup and Restore

Creating a Backup

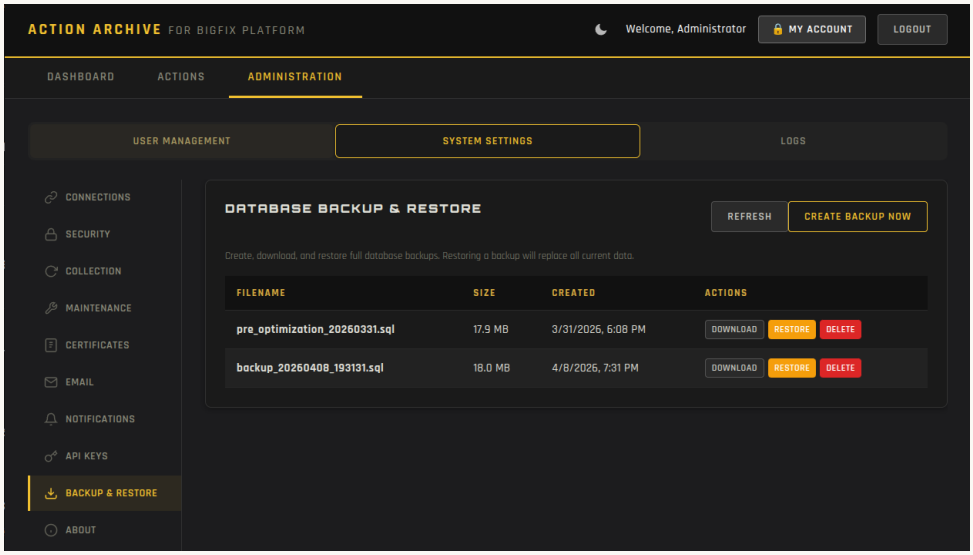
9. Click Create Backup Now.
10. The application runs pg_dump against the archive database.
11. The new backup file appears in the list with a timestamp and file size when complete.
12. Click the download icon to save the file to your local machine.

Restoring from a Backup

13. Click Restore next to the desired backup file, or upload a previously downloaded file.
14. Review the confirmation prompt carefully.
15. Click Confirm Restore.

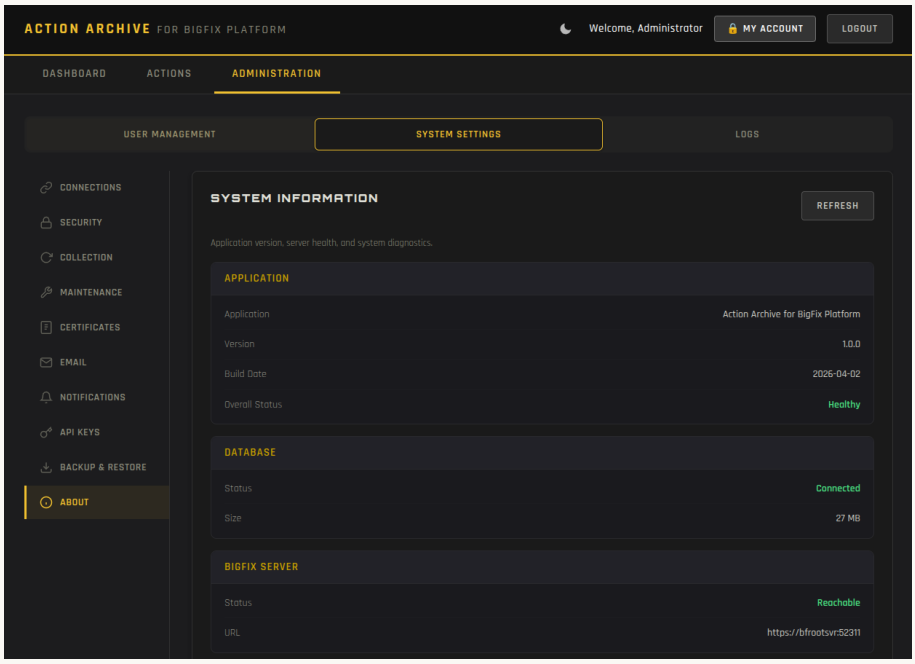
> ⚠️ **WARNING** Restoring a backup replaces ALL current database contents, including user accounts, system settings, archived actions, and audit logs. This operation cannot be undone. Always create a current backup before performing a restore.

Old backup files can be deleted from the interface to free disk space. Files are stored at /opt/bigfix-archive/backups/ on the server.



9.10 About

The About panel displays application metadata and live system health information: version and build date, server uptime, disk usage, memory usage, database size, and integrity verification status (see Section 12.5).

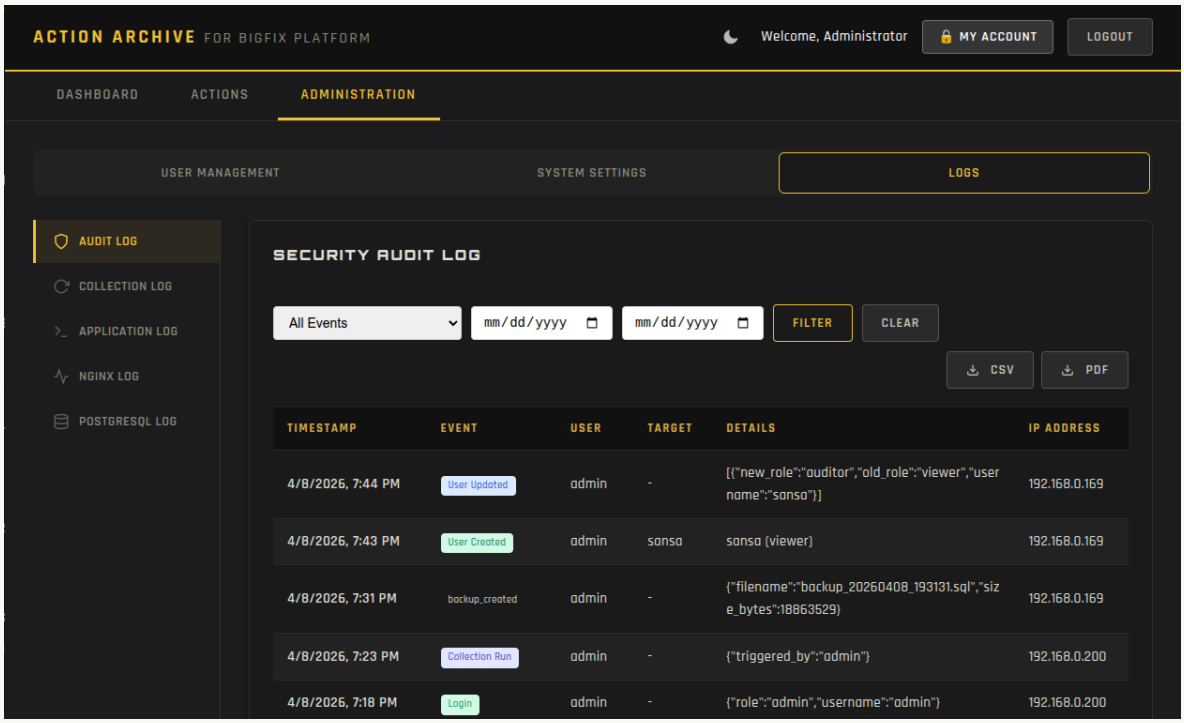


10. Administration — Logs

10.1 Audit Log

The Audit Log records every security-relevant event in the application and is available to Admin and Auditor roles from Administration → Logs → Audit Log.

Event Type	What Triggers It
LOGIN_SUCCESS	Successful user authentication.
LOGIN_FAILURE	Failed login attempt (wrong password or unknown user).
LOGOUT	User-initiated or session-expiry logout.
ACCOUNT_LOCKED	Account locked after exceeding max failed attempts.
ACCOUNT_UNLOCKED	Admin manually unlocked a locked account.
USER_CREATED	New user account created.
USER_UPDATED	User account details modified.
USER_DELETED	User account permanently deleted.
PASSWORD_RESET	Admin-initiated or self-service password reset.
SETTINGS_CHANGED	Any system setting modified.
COLLECTION_TRIGGERED	Manual or scheduled collection started.
COLLECTION_COMPLETED	Collection run finished successfully.
COLLECTION_FAILED	Collection run terminated with an error.
BACKUP_CREATED	Database backup file created.
BACKUP_RESTORED	Database backup restored.
PRUNING_RUN	Data pruning operation executed.
EXPORT_GENERATED	User generated a CSV or PDF export.
API_KEY_CREATED	New API key generated.
API_KEY_REVOKED	API key disabled or deleted.
CERTIFICATE_REPLACED	SSL certificate updated via the web UI.

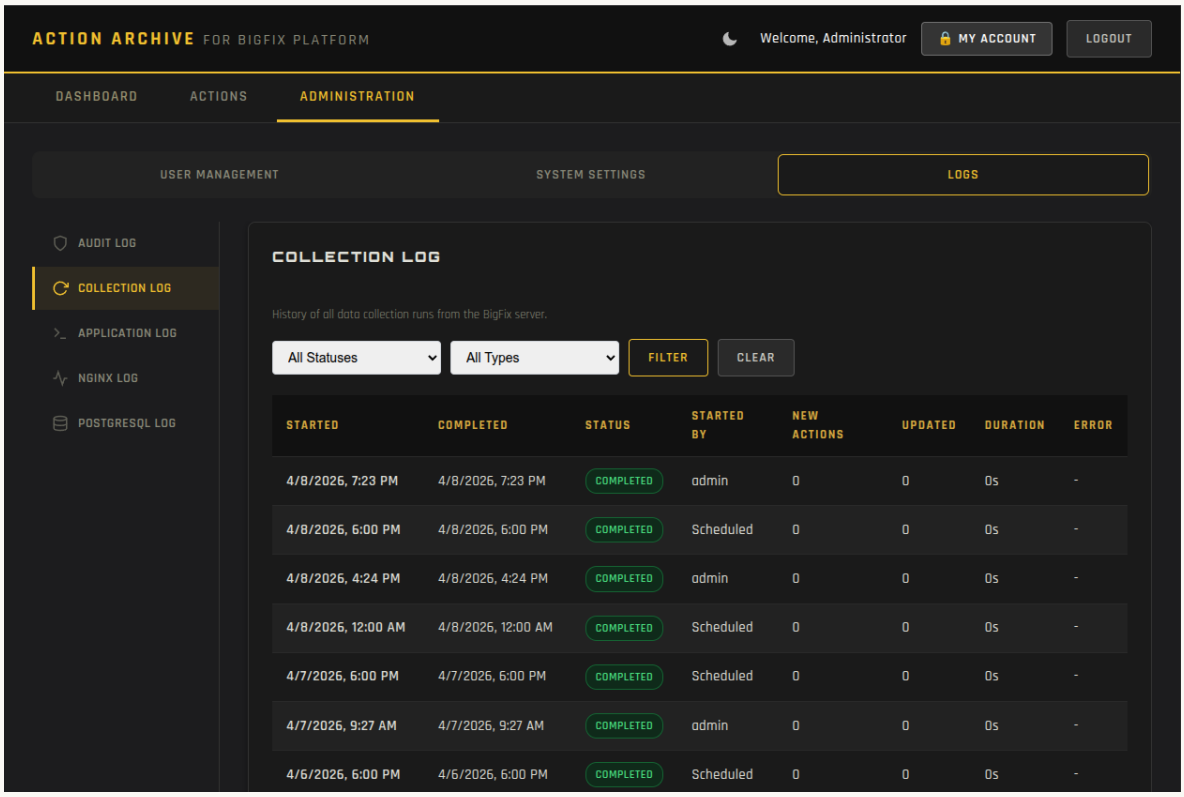


10.2 Collection Logs

Each collection run produces a detailed per-phase log accessed via the Details link in the Recent Collections table. The following phases are recorded:

Phase	Description
Initializing	Establishing connection to the BigFix REST API and Session Relevance endpoint. Verifying credentials.
Discovery	Running Session Relevance query to enumerate all expired and stopped actions across all sites.
Filtering	Comparing discovered action IDs against the database to identify new actions not yet archived.
Metadata	Bulk-fetching supplemental data: site names, operator display names, and parent MAG IDs.
Fetching	Retrieving full action XML and target-device results for each new action via the REST API.
Storing	Inserting or upserting action records, components, and target results into PostgreSQL.
Complete	Collection finished. Summary of actions archived and elapsed time logged.

Status	Meaning
Completed	The collection ran without errors and all discovered actions were stored.
Running	A collection is currently in progress.
Error	The collection failed. Review the per-phase log for the specific error.



10.3 Application, Nginx, and PostgreSQL Logs

- Application Log: Output from the FastAPI / Uvicorn process. Diagnose application errors, API exceptions, or startup failures. Equivalent to: journalctl -u bigfix-archive
- Nginx Access Log: All HTTP requests processed by Nginx, including response codes and sizes. Path: /var/log/nginx/bigfix-archive-access.log
- Nginx Error Log: Nginx-level errors, including upstream connection failures. Path: /var/log/nginx/bigfix-archive-error.log
- PostgreSQL Log: Database-level events, query errors, and connection issues.

11. Audit Agent Administration

11.1 Overview

The Action Archive Audit Agent is a lightweight Python utility that runs on the BigFix Root Server (Windows) as a Windows Scheduled Task. It monitors the BigFix server_audit.log file for action deletion events and reports them to the Archive REST API, preserving a record of actions deleted before they could be archived.

11.2 Configuration

The agent is configured via config.ini in its installation directory. Edits take effect on the next scheduled run — no restart is required.

Setting	Description	Example / Default
api_url	Full HTTPS URL of the Archive Server.	https://192.168.0.40
api_key	API key generated in Administration → Settings → API Keys.	aa-xxxxxxxxxxxx
audit_log_path	Full path to the BigFix server_audit.log file.	Auto-detected
position_file	File tracking the agent's last-read position in the audit log.	position.json
log_file	Path to the agent's own operational log.	agent.log
log_level	Logging verbosity: DEBUG, INFO, WARNING, or ERROR.	INFO

Log level guidance: DEBUG produces verbose output for troubleshooting; INFO (default) records each successful submission; WARNING captures non-fatal issues; ERROR records failures requiring attention.

11.3 Managing the Scheduled Task

16. Open Task Scheduler on the BigFix Root Server.
17. Locate Action Archive Audit Agent in the Task Scheduler Library.
18. Right-click to Run (manual trigger), Disable, Enable, or modify the trigger interval.

To uninstall, run the installer batch file and choose the Uninstall option. This removes the scheduled task but does not delete the installation directory or log files.

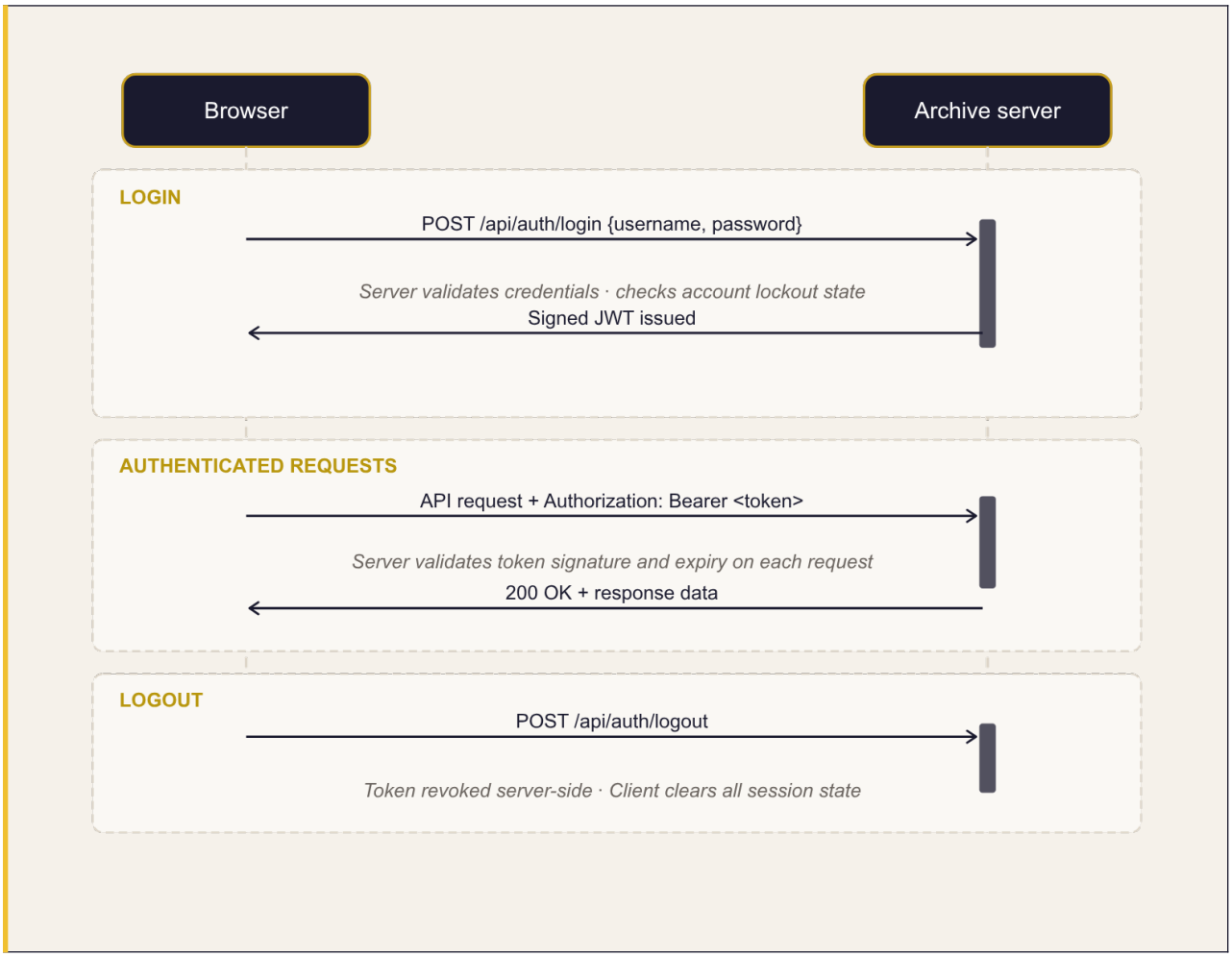
11.4 Troubleshooting the Audit Agent

Symptom	Likely Cause	Resolution
SSL certificate error	Archive Server uses self-signed cert.	Set verify_ssl = false in config.ini, or install the archive CA certificate into the Windows trust store.
API key rejected (401)	Key was revoked or copied incorrectly.	Generate a new API key in the archive and update api_key in config.ini.
Audit log path error	server_audit.log not found at configured path.	Locate the correct path in the BigFix installation directory and update audit_log_path.
Connection refused / timeout	Archive Server unreachable on port 443.	Verify the firewall allows the BigFix server to reach the archive on port 443.
No new records submitted	Position file is ahead of the current log.	Delete position.json to force a full re-read of the audit log from the beginning.

12. Security Reference

12.1 Authentication Model

Action Archive uses local authentication with JSON Web Tokens (JWT). There is no Active Directory or LDAP integration in this version.



Tokens are revoked server-side on logout. A revoked token list is maintained in the database for the duration of the token's original validity period, preventing token reuse.

12.2 Password Self-Service (Forgot Password)

When SMTP is configured, users can reset their own password using the Forgot Password link on the login page. Reset tokens are stored as SHA-256 hashes; the plaintext appears only in the email link. Tokens expire after 15 minutes and are invalidated immediately after a single use.

12.3 Credential Encryption

Sensitive values in the application configuration are encrypted at rest using a machine-derived key that is not written to disk. Encrypted values include: the BigFix operator password, PostgreSQL connection URL, application secret key, and SMTP password.

> ❗ **IMPORTANT** Encrypted credentials are machine-specific. If the application is migrated to a different server, all credentials must be re-entered through the web interface or the installer, as values encrypted on the original machine cannot be decrypted elsewhere.

12.4 Infrastructure Security Summary

Security Control	Implementation
Transport Security	TLS 1.2 and 1.3 only. Weak ciphers disabled. HSTS header enforced.
Reverse Proxy Isolation	Nginx terminates all TLS. FastAPI binds to localhost:5000 only and is not reachable externally.
Security Headers	HSTS, X-Frame-Options: DENY, X-Content-Type-Options: nosniff, X-XSS-Protection, Referrer-Policy.
CORS Policy	Cross-origin requests are disabled. The API only accepts requests from the same origin.
API Documentation	Swagger UI, ReDoc, and OpenAPI schema endpoints are disabled in production builds.
File Permissions	.env and manifest files are mode 600 (owner read/write only). Routers are mode 640.
Bytecode Prevention	PYTHONDONTWRITEBYTECODE=1 prevents compiled .pyc files from being written to disk.
Rate Limiting	IP-based login lockout after configurable failed attempts (Section 9.2).
Audit Trail	All user actions logged with timestamp, source IP, and user agent (Section 10.1).

12.5 Integrity Verification

Action Archive maintains a manifest of expected cryptographic hashes for all application files (.manifest.json). At startup and before each collection, the application verifies that no files have been modified. If the check fails:

- Further collection runs are blocked.
- An integrity failure flag is shown on the About panel.
- The X-Content-Signature: FAILED header is included in API responses to alert monitoring tools.

> ⚠️ **WARNING** A failed integrity check may indicate an incomplete update or unauthorized file modification. Review application logs and compare file hashes manually before re-enabling collections.

13. Service Management Reference

13.1 Linux Service Commands

The application runs as a systemd service named bigfix-archive. Run the following from a root shell or with sudo:

```
# Check service status
sudo systemctl status bigfix-archive

# Start / stop / restart
sudo systemctl start bigfix-archive
sudo systemctl stop bigfix-archive
sudo systemctl restart bigfix-archive   # required after .env edits

# Enable auto-start on server boot
sudo systemctl enable bigfix-archive

# View recent application logs (last 100 lines)
sudo journalctl -u bigfix-archive -n 100

# Follow application logs in real time
sudo journalctl -u bigfix-archive -f

# Reload Nginx after manual certificate changes
sudo systemctl reload nginx
```

13.2 Key File Locations

Item	Path
Application Directory	/opt/bigfix-archive/
Configuration File	/opt/bigfix-archive/config/.env
Python Virtual Env	/opt/bigfix-archive/venv/
Static Frontend Files	/opt/bigfix-archive/static/
Application Manifest	/opt/bigfix-archive/.manifest.json
Backup Directory	/opt/bigfix-archive/backups/
SSL Certificate	/etc/ssl/certs/bigfix-archive.crt
SSL Private Key	/etc/ssl/private/bigfix-archive.key
Nginx Site Config	/etc/nginx/sites-available/bigfix-archive

Item	Path
Systemd Service Unit	/etc/systemd/system/bigfix-archive.service
Application Logs	journalctl -u bigfix-archive (no physical log file)
Nginx Access Log	/var/log/nginx/bigfix-archive-access.log
Nginx Error Log	/var/log/nginx/bigfix-archive-error.log
PostgreSQL Database	bigfix_archive (manage via psql or pgAdmin)

13.3 Changing Configuration

Most settings are managed through the web interface and take effect immediately. A service restart is required only when the .env file is edited directly:

```
sudo systemctl restart bigfix-archive
```

ℹ️ **NOTE** In normal operation you should not need to edit .env directly. Use the web interface for all settings changes. Direct .env edits are reserved for recovery scenarios or initial setup.

14. Troubleshooting

The following table covers the most common issues encountered when deploying or operating Action Archive.

Symptom	Likely Cause	Resolution
502 Bad Gateway	FastAPI application not running.	Run: sudo systemctl status bigfix-archive. If stopped, restart. Check journalctl -u bigfix-archive for errors.
Login page does not load (connection refused)	Nginx not running.	Run: sudo systemctl status nginx. Restart if stopped.
Collection returns 0 actions (expected number of actions to be collected is >0)	BigFix operator lacks Master Operator privileges.	Log in to the BigFix Console, verify the configured operator has the Master Operator role assigned.
SSL certificate warning in browser	Self-signed certificate in use.	Expected. Proceed through the browser warning. For a permanent fix, replace with a CA-signed cert (Section 9.5).
Login fails after correct password	Account locked from failed attempts.	Admin unlocks via User Management, or wait for Lockout Duration to expire.
"Account temporarily locked" message	Rate limiter triggered.	Wait for lockout duration. Consider increasing the attempt threshold if users are frequently locked.
Cannot reach BigFix server	Firewall rule or routing issue.	Test: curl -sk https://<bigfix-server>:52311. Check firewall rules between the archive and BigFix Root Server.
Password reset email not received	SMTP not configured or misconfigured.	Check Settings → Email. Use Send Test Email to verify. Also check recipient spam folder.
Collection is slow on first run	Large number of actions fetched for the first time.	Expected. First run may take several minutes. Subsequent runs are incremental and significantly faster.
Audit Agent "connection refused"	Archive URL incorrect or port 443 blocked.	Verify api_url in config.ini. Test: curl -sk https://<archive-server> from the BigFix server.
Audit Agent "API key rejected" (401)	Key revoked or incorrectly copied.	Generate a new API key in the archive and update api_key in config.ini.
Integrity check failed (About page)	Application file modified or corrupted.	Review application logs, check for unauthorized changes, or reinstall from a known-good package.
Database size growing unexpectedly	Pruning not configured.	Enable scheduled pruning in Administration → Settings → Maintenance.
Backup download fails or is empty	Disk space exhausted on /opt partition.	Check: df -h. Delete old backup files in Administration → Settings → Backup & Restore.

Appendix A: API Quick Reference

The Archive REST API is available at https://<server>/api/. All endpoints require authentication via either a Bearer JWT token (from POST /api/auth/login) or an X-API-Key header. API documentation (Swagger UI) is disabled in production.

Method	Endpoint	Auth	Description
POST	/api/auth/login	None	Authenticate with username and password; returns a signed JWT.
POST	/api/auth/logout	JWT	Revoke the current JWT token server-side and end the session.
GET	/api/actions	JWT / Key	List archived actions. Supports filtering, sorting, and pagination.
GET	/api/actions/{id}	JWT / Key	Retrieve full detail for a single action.
GET	/api/actions/{id}/targets	JWT / Key	List all target devices and per-device status for an action.
GET	/api/actions/deleted	JWT / Key	List actions recorded as deleted before archiving.
POST	/api/collect	JWT	Trigger an on-demand collection run. Admin role only.
GET	/api/collection-runs	JWT	List recent collection runs with status and summary data.
GET	/api/collection-runs/{id}/logs	JWT	Retrieve per-phase log entries for a specific collection run.
GET	/api/users	JWT	List all user accounts. Admin only.
POST	/api/users	JWT	Create a new user account. Admin only.
GET	/api/audit-log	JWT	Retrieve audit log entries with filter support. Admin / Auditor.
GET	/api/settings	JWT	Retrieve current system settings. Admin only.
PUT	/api/settings	JWT	Update system settings. Admin only.
POST	/api/deleted-actions	Key	Submit a deletion event from the Audit Agent.
GET	/api/health	None	Lightweight health-check endpoint for monitoring systems.

ℹ️ **NOTE** Standard HTTP status codes apply: 200 Success, 401 Unauthenticated, 403 Insufficient role, 404 Not found, 422 Validation error, 500 Server error.

Appendix B: Data Storage Reference

The following table describes the categories of data stored by Action Archive at a conceptual level.

Data Category	What Is Stored
Actions	Action metadata, scripts, relevance expressions, issuer, status, dates, and fixlet references
Components	Child actions within Multi-Action Groups (MAGs) and Baselines, linked to their parent action
Targets	Per-computer status results for each action, including last report time and result messages
Users	User accounts, hashed passwords, roles, data-visibility scopes, and lockout state
Audit Trail	All user actions with timestamps, source IP addresses, user agents, and event details
Collection Runs	Collection history including start time, duration, status, and per-phase log entries
Deleted Actions	Action deletion events submitted by the Audit Agent before archival could occur
Configuration	All system settings stored as key-value pairs, managed through the web interface
API Keys	Hashed API key records for machine-to-machine authentication, with usage tracking

Database access is restricted to the application service account. Direct database connectivity from client machines is not permitted. For details on the backup and restore process, see Section 9.9.

Appendix C: Glossary

Term	Definition
Action	A deployment task created in BigFix that runs a script or applies a fixlet to one or more target devices.
Action Archive	This application. A web-based system for collecting, preserving, and querying stopped and expired BigFix action records.
Audit Agent	The Windows Scheduled Task component that monitors the BigFix server_audit.log and reports action deletions to the archive.
Baseline	A grouped BigFix action that deploys multiple fixlets as a single coordinated operation. Stored as a Multi-Action Group.
Bearer Token	A JWT credential included in the Authorization HTTP header to authenticate API requests.
BigFix Root Server	The central IBM BigFix server that manages policies, actions, and inventory for all managed endpoints.
Collection	A scheduled or manual process in which Action Archive queries BigFix for new expired / stopped actions and stores them.
Expired Action	A BigFix action whose end date or time has passed. Action Archive archives these actions when a collection runs.
Fixlet	A BigFix content item that defines a condition and an action to remediate it. Actions reference fixlets as their source.
Fixlet Site	A container in BigFix that groups related fixlets and actions (e.g., Patches for Windows, Custom Tasks).
Issuer	The BigFix operator username that created a given action.
JWT	JSON Web Token. A signed, time-limited credential used to authenticate web UI sessions.
MAG	Multi-Action Group. A parent BigFix action that coordinates a set of member (component) actions.
Master Operator (MO)	A BigFix operator with permissions to see and query all actions across all sites. Required for the archive collector account.
pg_dump	A PostgreSQL utility used to create portable database backups. Used by the archive's Backup & Restore feature.
Pruning	The process of permanently deleting archived action records older than the configured retention period.
Scope	A data-visibility restriction assigned to a user that limits which sites or issuers they can see in the archive.
Session Relevance	A BigFix query language for retrieving structured data from the BigFix server, used to supplement the REST API.
Stopped Action	A BigFix action manually stopped before its end date. Archived alongside

Term	Definition
	expired actions.
TLS	Transport Layer Security. The protocol used to encrypt HTTPS communications between browsers and the archive server.

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